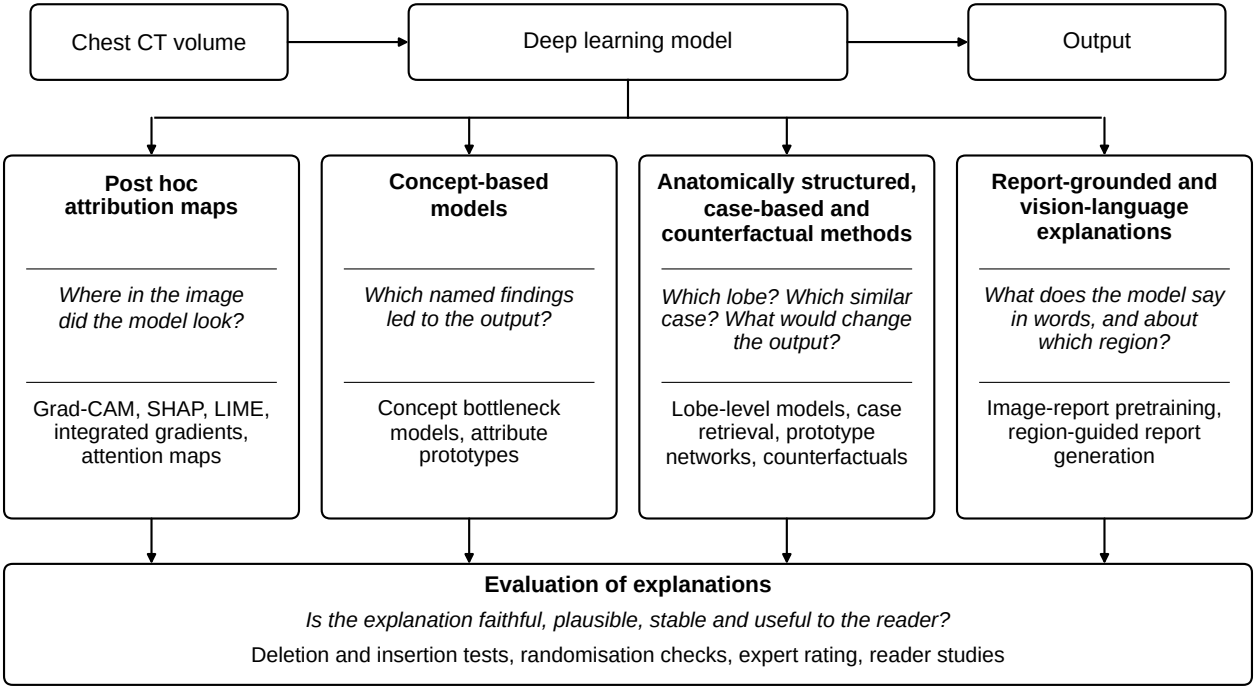
Chest CT volume
Deep learning model
Output
Post hoc attribution maps
Where in the image did the model look?
Grad-CAM, SHAP, LIME, integrated gradients, attention maps
Concept-based models
Which named findings led to the output?
Concept bottleneck models, attribute prototypes
Anatomically structured, case-based and counterfactual methods
Which lobe? Which similar case? What would change the output?
Lobe-level models, case retrieval, prototype networks, counterfactuals
Report-grounded and vision-language explanations
What does the model say in words, and about which region?
Image-report pretraining, region-guided report generation
Evaluation of explanations
Is the explanation faithful, plausible, stable and useful to the reader?
Deletion and insertion tests, randomisation checks, expert rating, reader studies